

PSET 3

①

(a) $\max U(c_1) + \beta(c_2)$

s.t:

$$\begin{cases} c_1 + a = a_0 + y_1 - \tau_1 \\ c_2 = y_2 + (1+r) \cdot a - \tau_2 \end{cases}$$

Replace c_2 for c_1 :

$$a = a_0 + y_1 - \tau_1 - c_1$$

$$c_2 = y_2 + (1+r) \cdot [a_0 + y_1 - \tau_1 - c_1] - \tau_2$$

$$U(c) = \frac{c^{1-\alpha}}{1-\alpha}$$

$$\max U(c_1) + \beta \cdot U(y_2 + (1+r)[a_0 + y_1 - \tau_1 - c_1] - \tau_2)$$

F.O.C:

$$U'(c_1) + \beta \cdot U'(y_2 + (1+r)[a_0 + y_1 - \tau_1 - c_1] - \tau_2) \cdot (1+r) \cdot (-1) = 0$$

$$U'(c_1) = \beta(1+r) \cdot U'(c_2) \rightarrow \left(\frac{c_2}{c_1} \right)^\alpha = \beta(1+r)$$

Euler
equation(s)

$$C_1 = C_2 \cdot [\beta(1+r)]^{-\frac{1}{\sigma}}$$

• From $t=2$ BC

$$C_2 = Y_2 - \tau_2 + (1+r)[a_0 + Y_1 - \tau_1 - C_2 \{[\beta(1+r)]^{-\frac{1}{\sigma}}\}]$$

$$C_2 (1 + \beta^{\frac{1}{\sigma}} (1+r)^{\frac{1}{\sigma}-1}) = Y_2 - \tau_2 + (1+r)[a_0 + Y_1 - \tau_1]$$

$$\therefore C_2 = \frac{Y_2 - \tau_2 + (1+r)[a_0 + Y_1 - \tau_1]}{1 + \beta^{\frac{1}{\sigma}} (1+r)^{\frac{1}{\sigma}-1}}$$

• From FOC

$$C_1 = \frac{a_0 + Y_1 - \tau_1 + \frac{Y_2 - \tau_2}{1+r}}{1 + \beta^{\frac{1}{\sigma}} (1+r)^{\frac{1}{\sigma}-1}}$$

• From Period 1 BC:

$$a = a_0 + Y_1 - \tau_1 - C_1 = \frac{\beta^{\frac{1}{\sigma}} (1+r)^{\frac{1}{\sigma}-1} (a_0 + Y_1 - \tau_1) - \frac{Y_2 - \tau_2}{1+r}}{1 + \beta^{\frac{1}{\sigma}} (1+r)^{\frac{1}{\sigma}-1}}$$

⑤

$$\frac{C_1}{Y_1} = \frac{a_0 + Y_1 - \tau_1 + \frac{Y_2 - \tau_2}{1+r}}{Y_1 (1 + \beta^{\frac{1}{\sigma}} (1+r)^{\frac{1}{\sigma}-1})}$$

$$\frac{\partial (C_1/Y_1)}{\partial Y_2} = Y_1 \cdot \beta^{\frac{1}{\sigma}} (1+r)^{\frac{1}{\sigma}-1} > 0$$

$$\uparrow \frac{C_1}{Y_1} \Rightarrow \downarrow a$$

⑥ $C_1 = \frac{a_0 + Y_1}{1 + \beta^{\frac{1}{\sigma}} (1+r)^{\frac{1}{\sigma}-1}} \rightarrow \frac{\partial C_1}{\partial r} = (a_0 + Y_1) \cdot \left(1 + \beta^{\frac{1}{\sigma}} (1+r)^{\frac{1}{\sigma}-1}\right)^{-2} \cdot \frac{\sigma-1}{\sigma}$

Reorganizing

$$\frac{\partial C_1}{\partial r} = \frac{(a_0 + Y_1) (1+r)^{\frac{1}{\sigma}-2}}{\left[1 + \beta^{\frac{1}{\sigma}} (1+r)^{\frac{1}{\sigma}-1}\right]^2} \cdot \left(\frac{\sigma-1}{\sigma}\right)$$

→ Determines
Sign of $\frac{\partial C_1}{\partial r}$

σ { Intertemporal
elasticity of
substitution

④ From the F.O.C:

$$C_1 = \frac{a_0 + Y_1 - T_1 + \frac{Y_2 - T_2}{1+r}}{1 + \beta^{1/\sigma} (1+r)^{1/\sigma - 1}}$$

$$\frac{\partial C_1}{\partial T_1} = \frac{-1}{1 + \beta^{1/\sigma} (1+r)^{1/\sigma - 1}}$$

$$\frac{\partial C_1}{\partial T_2} = \frac{\frac{-1}{1+r}}{(1 + \beta^{1/\sigma} (1+r)^{1/\sigma - 1})}$$

$$C_2 = \frac{Y_2 - T_2 + (1+r)[a_0 + Y_1 - T_1]}{1 + \beta^{1/\sigma} (1+r)^{1/\sigma - 1}}$$

$$\frac{\partial C_2}{\partial T_1} = \frac{-(1+r)}{1 + \beta^{1/\sigma} (1+r)^{1/\sigma - 1}}$$

$$\frac{\partial C_2}{\partial T_2} = \frac{-1}{1 + \beta^{1/\sigma} (1+r)^{1/\sigma - 1}} = \frac{\partial C_1}{\partial T_1}$$

↳ $\neq 0$ due to savings!

$$a = \frac{\beta^{1/\sigma} (1+r)^{1/\sigma-1} (a_0 + y_1 - \tau_1) - \frac{y_2 - \tau_2}{1+r}}{1 + \beta^{1/\sigma} (1+r)^{1/\sigma-1}}$$

$$\frac{\partial a}{\partial \tau_1} = \frac{-1 \cdot \beta^{1/\sigma} (1+r)^{1/\sigma-1}}{1 + \beta^{1/\sigma} (1+r)^{1/\sigma-1}} \rightarrow \underline{\in (-1, 0)}$$

$$\frac{\partial a}{\partial \tau_2} = \frac{\left(\frac{1}{1+r}\right) \cdot \beta^{1/\sigma} (1+r)^{1/\sigma-1}}{1 + \beta^{1/\sigma} (1+r)^{1/\sigma-1}} > 0$$

Saving extra to pay future taxes

③ $a = \frac{1}{1+\beta} \left[\beta(a_0 + y_1) - \frac{y_2}{1+r} - \left(\tau_1 + \frac{\tau_2}{1+r} \right) \right]$

→ PV of taxes

$$C_1 = \frac{a_0 + y_1 - \tau_1 + \frac{y_2 - \tau_2}{1+r}}{1 + \beta^{1/\sigma} (1+r)^{1/\sigma-1}}$$

$$C_2 = \frac{\frac{y_2}{1+r} + (a_0 + y_1) - \left(\tau_1 + \frac{\tau_2}{1+r} \right)}{1 + \beta^{1/\sigma} (1+r)^{1/\sigma-1}}$$

Just multiplied by $\frac{1+r}{1+r}$ what we had before

②

① $a \geq -b \leadsto$ Borrowing limit of $\underline{b}$

If $b=0 \leadsto$ No borrowing allowed

$$c_2 = y_2 - \tau_2 + (1+r) \cdot a \rightarrow b \geq \frac{y_2 - \tau_2 - c_2}{1+r}$$

Natural debt limit (implicit)

② Binds when household wants to borrow more. $\rightarrow$ High $\frac{y_2}{y_1}$
 $\rightarrow$ Low β

③ Non-binding case is ex. 1 with $a_0 = 0$

• When it binds:

$$\max U(c_1) + \beta \cdot U(c_2)$$

$$\text{s.t. } \underline{a = -b}$$

$$c_1 + a = y_1 - \tau_1$$

$$\rightarrow \underline{c_1 = y_1 - \tau_1 + b}$$

$$c_2 = y_2 - \tau_2 + (1+r) \cdot a \rightarrow \underline{c_2 = y_2 - \tau_2 - (1+r) \cdot b}$$

$$\text{Euler: } U'(c_1) > \beta(1+r) \cdot U'(c_2)$$

e) Different BC:

$$\max U(c_1) + \beta \cdot U(c_2)$$

$$\text{s.t: } \begin{cases} c_1 + a = Y_1 - \tau_1 \\ c_2 = Y_2 + (1+r-\tau_2) \cdot a \end{cases} \quad c_1 + \frac{c_2 - Y_2}{1+r-\tau_2} = Y_1 - \tau_1$$

• Solution close to before:

$$\frac{c_2}{c_1} = [\beta (1+r-\tau_2)]^{\frac{1}{\sigma}}$$

↳ Lowers $\frac{c_2}{c_1}$

$$c_1 = \frac{Y_1 - \tau_1 + \frac{Y_2}{1+r-\tau_2}}{1 + \beta^{\frac{1}{\sigma}} (1+r-\tau_2)^{\frac{1}{\sigma}} - 1}$$

↳ before: $\frac{Y_2 - \tau_2}{1+r}$

→ Same revenue → Both allocations are affordable
↳ Lump-sum is welfare improving → Euler equation